

**PHASE EQUILIBRIA AND CHEMICAL PROCESSES
IN SYSTEMS ZnO-SiO₂-NiO AND ZnO-SiO₂-MnO**

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This paper examines chemical interactions and phase equilibria in the ternary systems ZnO-SiO₂-NiO and ZnO-SiO₂-MnO. This information is necessary for selecting optimal conditions for the synthesis of functional materials based on willemite Zn₂SiO₄ (green phosphor Zn₂SiO₄:Mn, cool blue pigment Zn_{2-2x}Ni_{2-2x}SiO₄) and olivine Ni₂SiO₄. Solid-phase reactions between three reactants begin at the interfaces of only two at a time. In the ZnO-SiO₂-NiO and ZnO-SiO₂-MnO systems synthesized in air, the first pair of oxides to react are metal oxides, forming solid solutions of Ni_{1-x}Zn_xO (rock salt structure) and Zn_{1-x}Mn_xMn₂O₄ (spinel structure), respectively. With further increase in annealing time, the proportion of the dopant ion in the solid solutions increases. Upon heating to 900 °C, zinc and silicon oxides begin to react in both systems, forming willemite Zn₂SiO₄, which gradually becomes enriched with dopant metal ions to form the solid solution Zn_{2-2x}M_{2-2x}SiO₄ (M = Ni or Mn).

At higher temperatures, the reaction of Ni_{1-x}Zn_xO and SiO₂ in the ZnO-SiO₂-NiO system results in the formation of a phase with the olivine structure Ni_{2-2x}Zn_{2-2x}SiO₄. Phase equilibria in the system are determined by the quasi-binary equilibrium of the boundary compositions of the Zn_{1.8}Ni_{0.2}SiO₄ and Ni_{1.5}Zn_{0.5}SiO₄ solid solutions and the conodes connecting these compositions with the Ni_{0.83}Zn_{0.17}O solid solution, as well as the conode connecting the boundary composition of the Ni_{1.6}Zn_{0.4}O solid solution with Zn₂SiO₄.

In the ZnO-SiO₂-MnO system, no phases with an olivine structure were detected. In the temperature range of 800-1250 °C, the sequential appearance of intermediate phases ZnMnO₃, ZnMn₂O₄, and MnSiO₃ was detected. Phase equilibria were established for temperatures of 800 °C and 1250 °C. It is shown that the phase relationships in the MnOx-ZnO-SiO₂ system are determined by changes in the charge states of manganese ions with increasing temperature, leading to phase transformations of manganese oxide MnO→Mn₂O₃→Mn₃O₄→MnO. Triangulation of the Mn₂O₃-ZnO-SiO₂ system at 800°C is determined by the conode ZnMn₂O₄-Zn₂SiO₄ and decomposes the system into elementary triangles ZnO-Zn₂SiO₄-ZnMn₂O₄, Zn₂SiO₄-ZnMn₂O₄-SiO₂, and ZnMn₂O₄-SiO₂-Mn₂O₃. The triangulation of the ternary system MnO-ZnO-SiO₂ is determined by the elementary triangle Zn_{1.6}Mn_{0.4}SiO₄-ZnO-MnSiO₃.

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